

Problem 6

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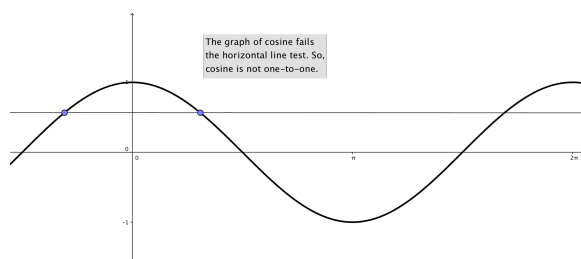
Problem 1 Without using a calculator, determine if the statement

$$\cos^{-1}(\cos(7\pi/6)) = 7\pi/6$$

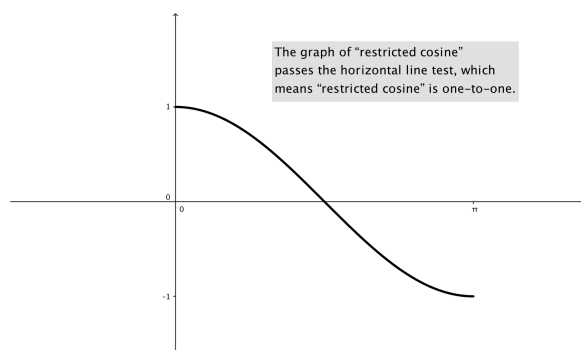
is true or false.

Explanation. This statement is **false**: the correct statement is $\cos^{-1}(\cos(7\pi/6)) = 5\pi/6$. (Why?)

Spoiler Alert: the cosine function is *not* invertible since its graph fails the horizontal line test.



To produce the inverse cosine we must first restrict the domain of cosine, to the interval $[0, \pi]$, to produce an invertible function:



So, the range of $\cos^{-1}$ is $[0, \pi]$. Since $7\pi/6$ is not in this range, $7\pi/6$ is *never* a possible output of $\cos^{-1}$.